DE–8354 **11**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2014.

MATHEMATICAL FOUNDATION OF COMPUTER SCIENCE

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Write the truth table for $Q \to P$.

2. Let $p(x)$ denote the statement *"x > 4"*. What are the truth values of *p(5)* and *p(2)*.

3. What are finite and infinite sets? Give an example.

4. Define Equivalence relations.

5. What is bijective? Give an example.

6. Define composition of functions.

7. What is submonoid?

8. Define abelian group.

9. What is cycle? Give an example.

10. What is path? Give an example.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain any three connectives with examples.

12. Show that $R \wedge (PVQ)$ is valid conclusion form the premises $PVQ, Q \to R, P \to M$ and $\rceil M$.

13. Given $S = \{1,2,3,....10\}$ and a relation R on S where $R = \{(x, y) / x + y = 10\}$ what are the properties of the relation R?

14. Using characteristics function show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

15. Show that the set N of natural numbers is a semigroup under the operation $x * y = \max \{x, y\}$. Is it a monoid?

16. Show that the sum of the degree of the vertices of a graph G is twice the number of lines i.e $\Sigma \deg v_i = 2q$.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Show that P is equivalent to the following formulas

(a) $(P \wedge Q) V (P \wedge \neg Q)$

(b) $(P \vee Q) V (P \vee \neg Q)$

18. Prove that $\psi_{AUB}(X) = \Psi A(X) + \Psi_B(X) - \Psi_{A \cap B}$ where +, - are usual arithmetic operations.

19. State and prove the Lagrange's theorem.

DE–8355 **12**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2014.

COMPUTER ARCHITECTURE

(2007 Onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Give the importance of parallel processing.

2. Define multi computers.

3. What do you mean by program partitioning?

4. Mention the need for scalability analysis.

5. Define data arrays.

6. Define pipelining.

7. What do you mean by cache memory?

8. Define the term I/O channel.

9. What do you mean by multi processor?

10. Explain multi threading.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain about multivector and SIMD computers.

12. Discuss on VLSI models.

13. Give the importance of program scheduling

14. Discuss on the scalability of parallel algorithms.

15. Describe the environment needed for parallel programms.

16. Explain about the importance issues of multi processors.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Write about architectural classifications of parallel computers.

18. Explain the following:

    (a) Program flow mechanism

    (b) Network properties and routing.

19. Discuss in detail about virtual memory systems.

DE–8356 **13**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2014.

DATA STRUCTURES USING C++

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. How to declare two dimensional array?
2. Define Inheritance.
3. Mention the operations performed on STACK?
4. How Queue differs from a stack?
5. List the concept of Linked list.
6. Give an example for doubly linked list.
7. Define Graph.
8. How to insert an element in Binary tree?
9. What is meant by searching?
10. How time is measured in sorting?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Give the purpose of constructors.

12. How Queue can be expressed as an ADT?

13. Discuss the operations on Binary Trees.

14. Explain INORDER traversal and POST ORDER traversal.

15. Discuss on Quick sort.

16. How sequential search differs from Binary search?

SECTION C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. Write a program in C++ for operator overloading.

18. How to Insert and delete the elements in doubly linked lists?

19. Discuss on :

    (a) Selection sort

    (b) Merge sort.

DE–8357 **14**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2014.

VISUAL PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What is an event in windows programming?

2. What is an Interface?

3. Specify the difference between Command and Label in VB.

4. Mention any three properties of Grid control.

5. What is a use of data control in VB?

6. What is a bit map?

7. What is an object?

8. Expand ODBC of DLL.

9. What is MDI?

10. Give any three DAO classes in VC++.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the following :

    (a) Windows programming Environment

    (b) Data types in VB.

12. Explain SELECT..CASE and SWITCH with proper examples.

13. Explain about Menu creation in VB.

14. Explain the steps in creating a dialog box in VC++.

15. Explain about OLE.

16. What are the constituents of a DLL? How it is different from a runtime library? What are its strength?

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Write in detail about procedures and function in VB.

18. Design and implement employee data base in VB.

19. (a) Explain in detail about serialization in VC++.

    (b) Explain the use and purpose of MDI in VC++.

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DE–8358 **15**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2014.

DATA BASE MANAGEMENT SYSTEMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Define database.

2. What is meant by schema?

3. Mention any two semantic issues related in relational model.

4. Define Relational model.

5. Name any three types of Databases.

6. List any three benefits of Oracle.

7. How objects are defined in Data Base?

8. What is meant by Union?

9. Define Table.

10. Give the expansion for SQL.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Mention the salient features of Data Base Management System.

12. How to prepare the subschema for the database?

13. List the concept of Relational Model.

14. Discuss the network system followed in Oracle.

15. Discuss on any five Data Access SQL commands.

16. How to rename a table?

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain the different user Interfaces in Data Base Management System.

18. Describe the architecture of Oracle database.

19. How to join multiple tables in a Query? Explain with examples.

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DE–8359 **21**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2014.

COMPUTER NETWORKS

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Write notes on applications of computer networks.

2. Define physical layer and network layer.

3. Discuss about one bit sliding window protocol.

4. Write about HDLC.

5. Write short notes on RARP.

6. Define Routing.

7. Write a note on transport layer.

8. Define Gateway.

9. Define DNS.

10. Discuss about E-mail.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain Telephone system in detail.

12. Discuss about PPP and ALOHA.

13. Write a detail note on congestion control algorithm.

14. Explain about UDP.

15. Discuss about client and server side.

16. Explain DNS in detail.

SECTION C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. Differentiate OSI and TCP reference model.

18. Discuss Routing algorithms.

19. Explain briefly about Audio.

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DE–8360 **22**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2014.

SOFTWARE ENGINEERING

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Define software engineering.

2. Give the purpose of the project metric.

3. How to estimate the project?

4. What is meant by risk mitigation?

5. How to define a task set for the software project?

6. Define software reliability.

7. How to analysis the software engineering concepts?

8. Give the importance of data design.

9. What is meant by validation?

10. Why debugging is said to be the art?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Discuss on the linear sequential model.

12. Mention the concepts of project management.

13. What are the techniques involved in decomposition?

14. Discuss on ISO 9000 quality standards.

15. Explain the concept in transform mapping design.

16. Discuss in White Box testing.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain the following models
    (a) Proto typing model
    (b) Linear sequential model.

18. How to conduct statistical tests for quality assurance?

19. Describe the architectural design in software engineering.

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DE–8361 **23**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2014.

INTERNET PROGRAMMING AND WEB DESIGN

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)
Answer ALL questions.

1. Explain briefly about WWW.
2. What do you mean by PERL?
3. List out the various data types in JavaScript.
4. State any two key features of VBScript.
5. Explain SGML in brief.
6. Sate any three key animation features of Shockwave.
7. Explain about Plug-Ins.
8. Write about multiple protocols.
9. Explain briefly about Animation.
10. What is an ActiveX control?

SECTION B — (4 × 10 = 40 marks)
Answer any FOUR questions.

11. What is Internet? Explain how to develop internet applications.

12. Write a VB Script to find the following

    (a) greatest among three values

    (b) factorial of a given number.

13. Write short notes on the following:

    (a) Netscape navigator

    (b) Internet Explorer.

14. Explain how to create an ActiveX control to activate a Webpage.

15. Explicate how to create a custom integrated application.

16. Describe in detail about images and hyperlinks.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain in detail Encryption and Decryption with an example.

18. Create a College website using HTML tags.

19. Discuss in detail about the formatting text in HTML with examples.

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DE–8362 **24**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2014.

IMAGE PROCESSING AND ANALYSIS

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What is the difference between intensity and brightness?
2. What do you mean by camera calibration?
3. What is the use of smoothing operation?
4. What are the categories of degradation?
5. Classify the error criteria.
6. Write the steps that are used in the process of mapping.
7. What are the two important properties of segmentation?
8. What are the criteria considered by the canny for localizing edges?
9. When a pixel is said to be critical pixel?
10. What is intra-class distance?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Describe the components of image processing system.
12. Describe the basic transformations of geometric model.
13. Explain the image sharpening and crispening using Laplacian.

14. Describe the various steps in canonical frame registration.

15. With an example, explain the region splitting and merging technique.

16. How can you obtain the geometrical attributes?

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain the method of constrained least-square estimation.

18. Explain the following coding methods with suitable examples.

    (a) Run-length coding

    (b) Block coding.

19. (a) Describe the Canny edge detection algorithm.

    (b) What are the ways available to represent a boundary? Explain any two methods.

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DE–8363 **25**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2014.

OPERATING SYSTEMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What are the advantages of Multiprogramming?

2. List out the performance metrices of CPU scheduling algorithms.

3. What is the purpose of resource allocation graph?

4. Define the semaphore primitives.

5. What do you mean by swapping?

6. Why page size is always power of 2?

7. What are the criteria to be consider in choosing a file organization?

8. What do you mean by access control matrix?

9. List out the Naming schemes.

10. Write the disadvantages of single-coordinator approach.

SECTION B — ($4 \times 10 = 40$ marks)

Answer any FOUR questions.

11. Explain the layered approach of OS.

12. With an example, explain the structure of thread.

13. Describe the paging scheme of memory management.

14. Explain how the free-space list is maintained.

15. Describe the fully distributed approach for implementing mutual exclusion in a distributed environment.

16. How can you recover from deadlock?

SECTION C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. Discuss the various issues of caching in a remote-service mechanism.

18. Describe the various disk scheduling algorithms.

19. What do you mean by page-fault? Explain any two page replacement algorithms.

DE–8827 **17**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION, MAY 2014.

COMPUTER LAB II — VISUAL PROGRAMMING

(2007 onwards)

Time : Three Hours Maximum : 100 Marks

Examiner has to choose any ONE question.

1. (a) Write event procedure for the following:
   (i) Display date and time in the label box at run time
   (ii) Find the power of a number (Accept 2 inputs as number and power)
   (iii) Temperature conversion (from Fahrenheit to centigrade)

   (Use Label, text, command button in the form)

   (b) Write a VC++ program to find out whether a mouse is attached or not; and if attached how many buttons are present or not.

---------- Cut here ----------

2. (a) Write event procedure to perform the following:
   (i) Reverse a string
   (ii) Determine whether the given string is palindrome or not
   (iii) Change the case of a string to upper of lower

   (use label, text, command button)

   (b) Write a VC++ program to create a list box in a window.

3. (a) Using MS flex grid control, display the multiplication and addition table of 20 rows and columns (use MS Flex Grid, Button)

(b) Write a VC++ program to get the status of the shift and toggle keys using MFC.

---------- Cut here ----------

4. (a) Using built in Ax control, develop the windows NOTEPAD with file and edit menu operation and display the floating menu whenever necessary (use RTbox, Menu editors)

(b) Write a VC++ program to create a window of desired size using MFC.

---------- Cut here ----------

5. (a) Create a form with text box, combo box and command button and do the following

(i) Add the university name in the combo at the run time

(ii) Search and delete the particular university in the combo box.

(iii) Display the message box with number of university available in the combo box

(iv) Sort the university names in the alphabetical order.

(b) Write a VC++ program to get the status and toggle keys using MFC.

6. (a) Write a VB program to perform the following operation in a record of random access file.

   (i) Inset

   (ii) Delete

   (iii) Search

   (b) Write a VC++ program to create a window of desired size using MFC.

---------- Cut here ----------

7. (a) Using ActiveX control create a text box that accepts only numeric value with the following properties.

   (i) Background of the text box

   (ii) Foreground of the text box

   (iii) Text property of the text box

   (iv) Resize property of the text box.

   (b) Write visual C++ program to fill background of the client area with a bitmap.

---------- Cut here ----------

8. (a) Using a control Array, create a simple calculator which will do the following operations.

   (i) Addition

   (ii) Subtraction

   (iii) Multiplication

   (iv) Division

   (v) Percentage

   (vi) Square root

   (vii) Modulus

   (viii) Power.

   (using text box, command button, frames)

   (b) Write a VC++ program to create a list box in a window.

9. (a) Write a VB program to perform the following operation in a record of random access file.

   (i) Insert

   (ii) Delete

   (iii) Search.

   (b) Write a VC++ program to generate a status bar and show the status of caps lock. Num lock and scroll lock in it.

---------- Cut here ----------

10. (a) Create an employee with Empno, Empname, Basic pay, HRA, DA, PF, LIC, GP and NP with the following Calculation.

   HRA = 10% of BP

   PF = 3% of BP

   GP = BP+DA+HRA

   DA = 5% of BP

   LIC = 5% of BP

   NP = GP – (PF+LIC)

   Using remote data access object, implement the following operations.

   (i) Inset a record

   (ii) Search and delete a record

   (iii) Modify the record

   (iv) Display all the employee records whose names are starting with the letter 'S'.

   (b) Write a VC++ program to create a list box in a window.

DE–8826 **16**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2014.

COMPUTER LAB I — C++ AND DATA STRUCTURES LAB

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select and give ONE question to each candidate by lot system.

Break-Up of Marks

Record	:	10
Algorithm	:	10
Program	:	50
Debugging	:	10
Execution	:	10
Result	:	10
Total	:	100

1. (a) Write a C++ program to convert a given number into words for numbers 1 to 5.

   (b) Write a C++ program to sort a set of elements using selection sort.

---------- Cut here ----------

2. (a) Write a program to find the roots of a quadratic equation $ax^2 + bx + c = 0$.

   (b) Write a C++ program to sort a set of elements using Insertion sort.

3. (a) Write a C++ program to find the roots of a quadratic equation using function.

   (b) Write a C++ program to sort a set of elements using Quick sort.

---------- Cut here ----------

4. (a) Write a C++ program for matrix addition and matrix subtraction.

   (b) Write a C++ program to sort Ten numbers in ascending order.

---------- Cut here ----------

5. (a) Write a C++ program for Matrix Multiplication.

   (b) Write a C++ program to sort Ten numbers in descending order.

---------- Cut here ----------

6. (a) Write a C++ program to find the factorial of a given number using recursion.

   (b) Write a C++ program for linked list implementation of queue operations.

---------- Cut here ----------

7. (a) Write a C++ program to read 10 values to an array variable. Use pointers to locate and display each value.

   (b) Write a C++ program to implement insert and delete operations on queue using array concept.

8. (a) Using operator overloading write a C++ program to find the difference and total length of given two various tubes specified in meters and centimeters.

(b) Write a C++ program to implement push and pop operations on stack.

---------- Cut here ----------

9. (a) Using dynamic constructors write a C++ program to concatenate two given strings.

(b) Write a C++ program to evaluate the given mathematical expressions using stack.

---------- Cut here ----------

10. (a) Write a program using polymorphism to calculate the square of any two number of type int, Float, double and long.

(b) Write a C++ program to implement insert and delete operations on linked list structure.

DE–8828 **26**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2014.

INTERNET PROGRAMMING – LAB III

(2007 onwards)

Time : Three hours Maximum : 100 marks

Break-Up of Marks

Record	:	10
Algorithm	:	10
Program	:	50
Debugging	:	10
Execution	:	10
Result	:	10
Total	:	100

Examiner has to choose any ONE question.

1. (a) Design a web page using HTML to display your class time table.

   (b) Write a Java program using Applet to display any three images when three buttons in the Border layout are clicked. The image – should be displayed at the center.

2. (a) Write a menu driven Java Program to do database functions using database connectivity facility. The functions include display, edit, insert and delete a records from a table.

   (b) Design a web page for a car showroom to be opened shortly.

---------- Cut here ----------

3. (a) Write a Java Program using applet to display the dialogue menu in applet.

   (b) Write a VBScript to create a calendar for given month and year.

---------- Cut here ----------

4. (a) Write a Java Program using applet to create the frames and its controls.

   (b) Develop a web page for job recruitment agency in an IT industry.

---------- Cut here ----------

5. (a) Write a Java Script to create a color palet and display the text in the color choosen from the palet with proper background color.

   (b) Develop a picture gallery having atleast 5 pages. Each of them is having several pictures with suitable information.

6. (a) Write a JavaScript to create a order form to select the house articles.

   (b) Design and publish a personal web page.

---------- Cut here ----------

7. (a) Write a Java Script to create a window by using the confirm message.

   (b) Write a swing program to create the tabbed panels.

---------- Cut here ----------

8. (a) Write a VBScript to do the following :

   (i) Check the given password.

   (ii) Change the existing password.

   (b) Write a Java Program to create a color as the background.

---------- Cut here ----------

9. (a) Write a VBScript to do the following :

   (i) Display today's date

   (ii) Find the difference between 1/7/13 and 28/7/13

   (iii) Find the age of a person whose date of birth is 12/01/2000.

   (b) Write a Java Program using Applet to display the different colors and fonts.

10. (a) Write a swing program to create buttons with

    (i) Tool tip text

    (ii) Image

    (iii) Border

    (iv) Short cut key.

    (b) Create a web page in the format of front page of a news paper using Text Links. Align the text with colors.

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